香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

Lecture 21: Fluid

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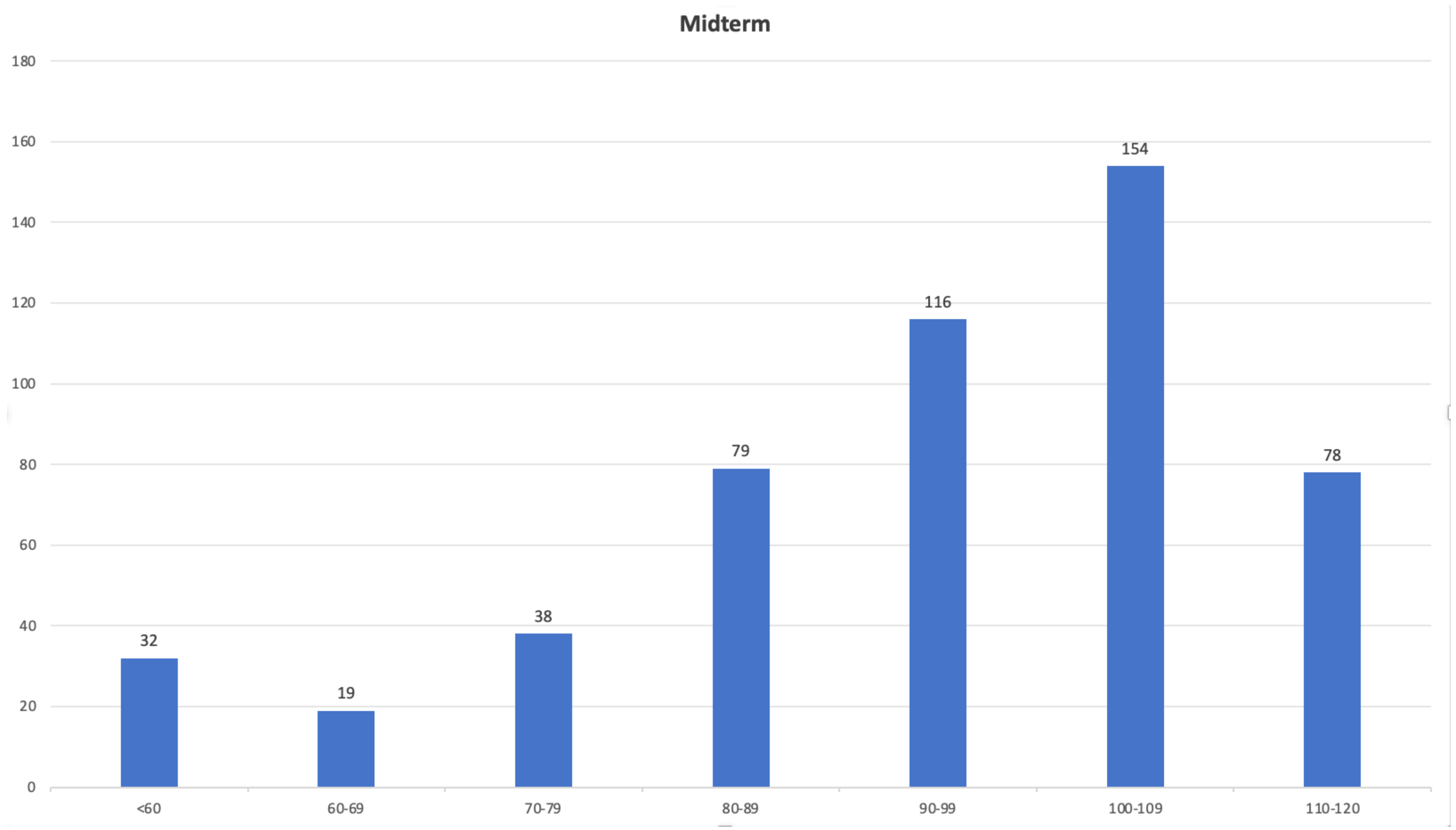
April 8th, 2025

Announcement

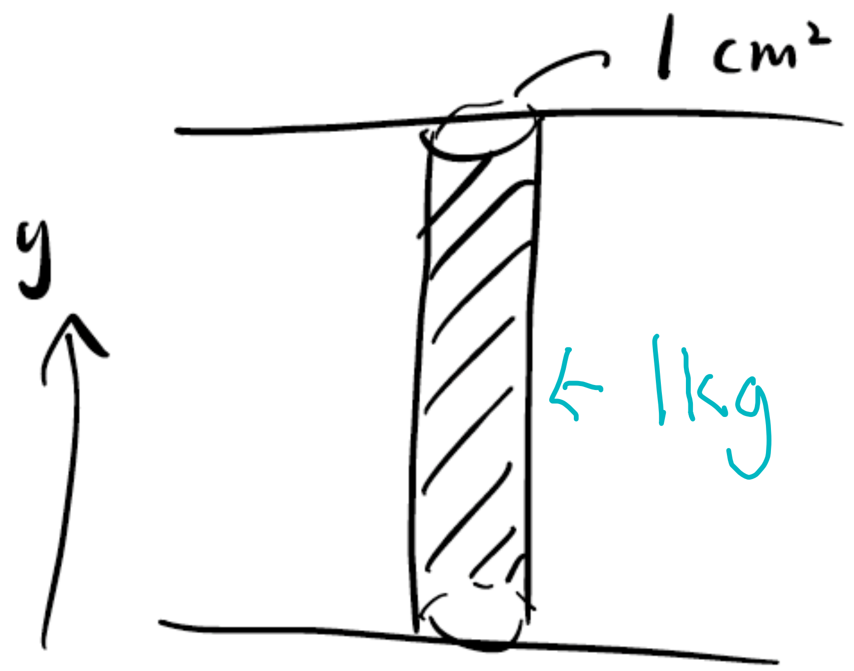
- Midterm grade available on BB
- Quiz on Thursday (April 10th)
  - About gravitation



Recap: Fluid

- Pascal's Principle: Pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid.
- Imcompressibility/continuity: $A_1 d_1 = A_2 d_2$
- Hydrostatic pressure: $-\rho g = \dfrac{dP}{dy}$
- Assume liquid is incompressible: $P_2 - P_1 = \rho g (y_1 - y_2)$

Atmospheric/Barometric Pressure



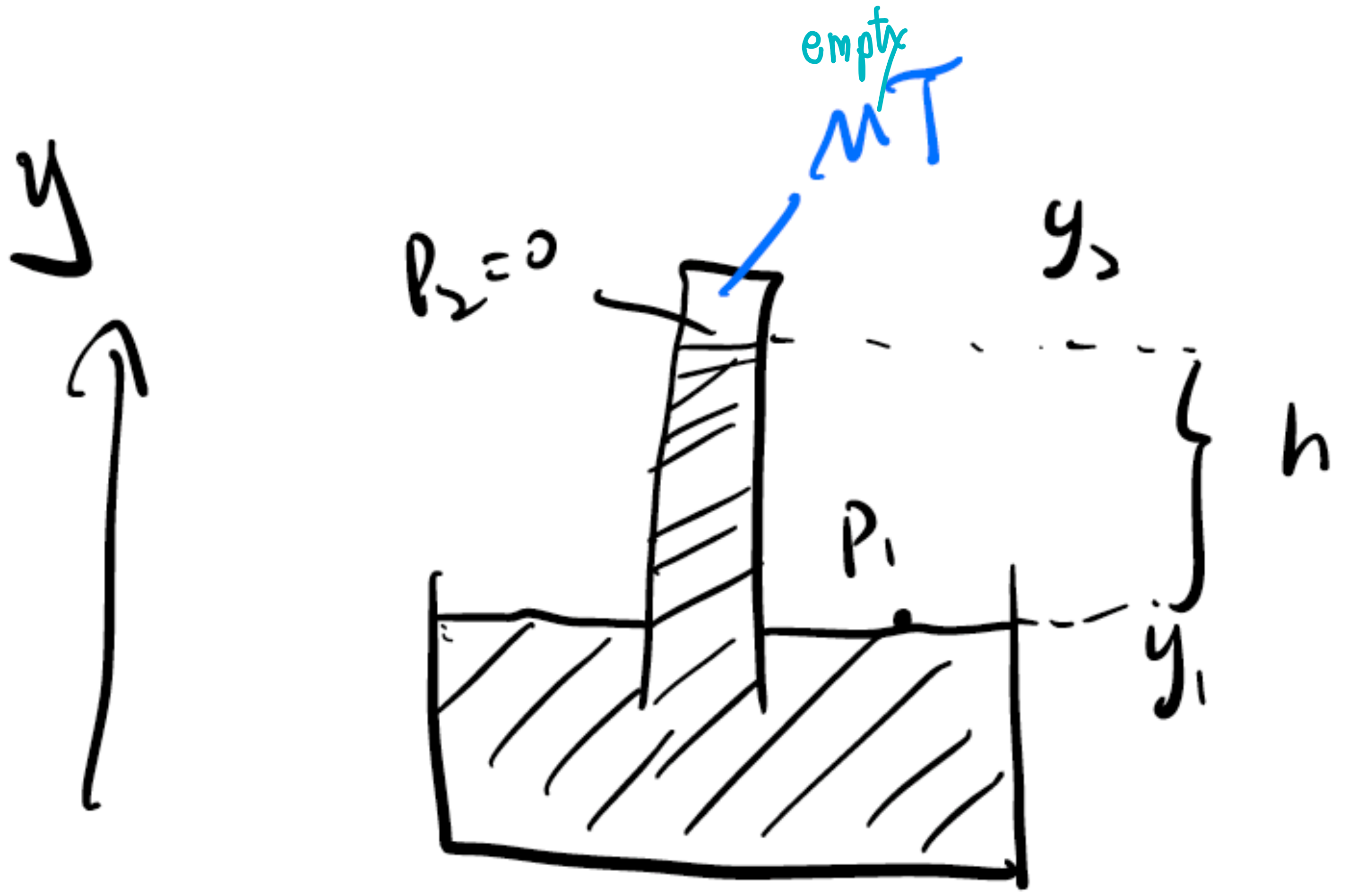
$F = 10\,N,\ 1\,cm^2$

$$P = \frac{F}{A} = \frac{10\,N}{1\times10^{-4}\,m^2} = 10^5\,N/m^2$$

$$\approx 10^5\,Pa$$

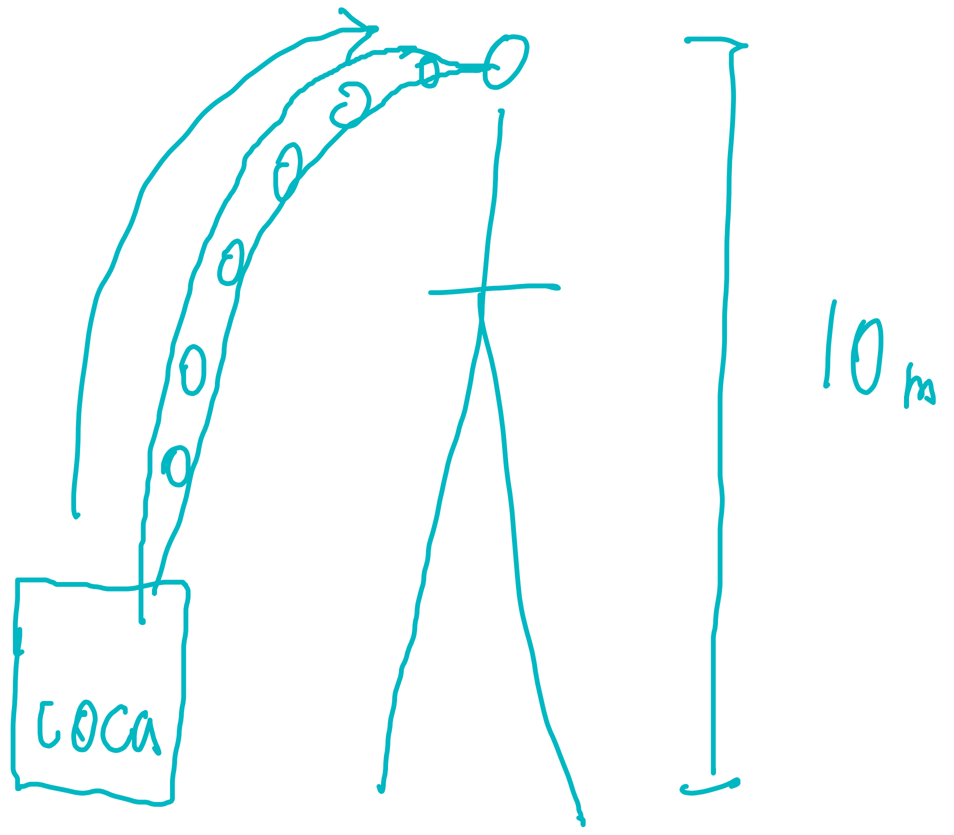
Atmospheric pressure = 1×10^5 Pa

How to Measure Atmospheric Pressure?



$$P_2 - P_1 = \rho g (y_1 - y_2)$$
$$= \rho g (-h)$$

$$P_1 - \underset{0}{P_2} = \rho g h$$



$P_{atmos} = 10^5$ Pa

$\rho_w = 10^3$ kg/m^3 $\quad h = 10$ m

$\rho_{Hg} = 13.6 \times 10^3$ kg/m

$h = 76$ cm $= 0.76$ m

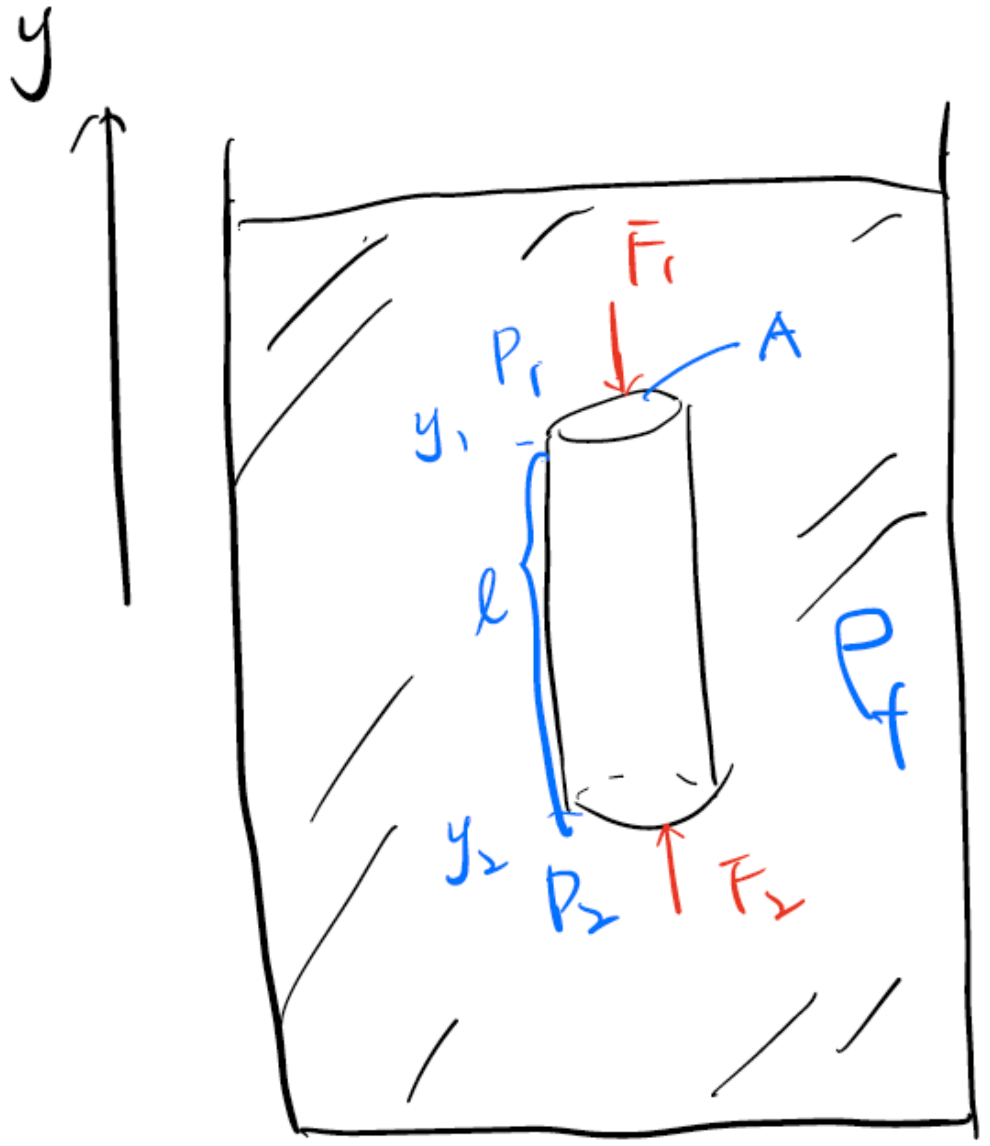
Buoyancy

in the liquid

(if full in liquid ⇒ Buoyancy)

(always upward)

(gas → hydrastatic) (gravity → down)



gravity acceleration

- $F_B = \rho_f g V$, where
  - F_B: buoyancy force
  - ρ_f: density of fluid
  - V: volume of the object in the fluid

$$P_2 - P_1 = \cancel{\rho g (y_1 - y_2)}$$

$$= \rho g l$$

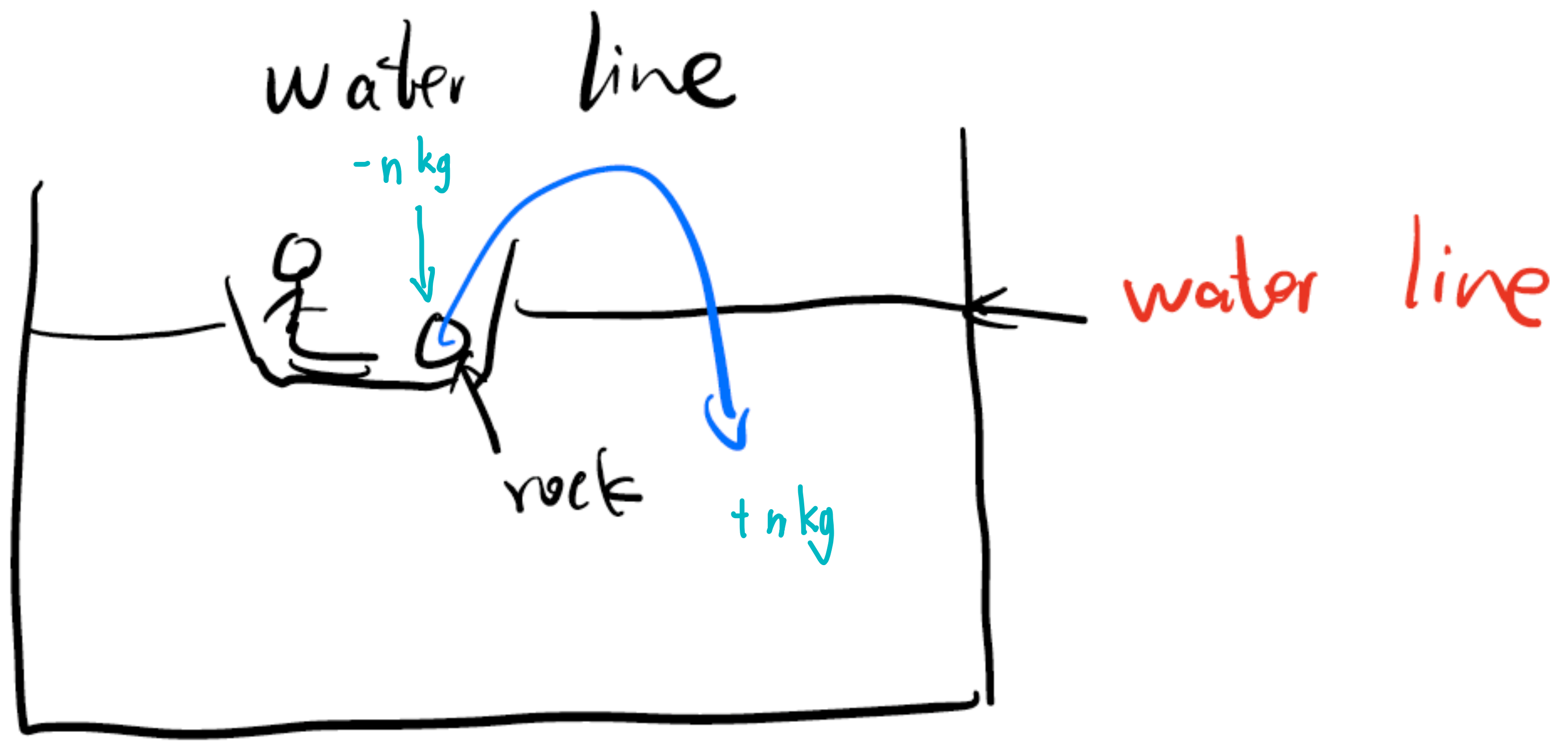
hydrastatic pressure

$$F_B = F_2 - F_2 = P_2 \cdot A - P_1 \cdot A$$

$$= \rho g l \cdot A$$

$$= \rho g V$$

Water Line



You throw rocks from a boat into water. Does the water line change?

$F_B = \rho g V_{dis}$

① F_B

mg

$F_B = mg$

$\rho_w \cdot g \cdot V_{dis} = \rho_R \cdot V_R \cdot g$

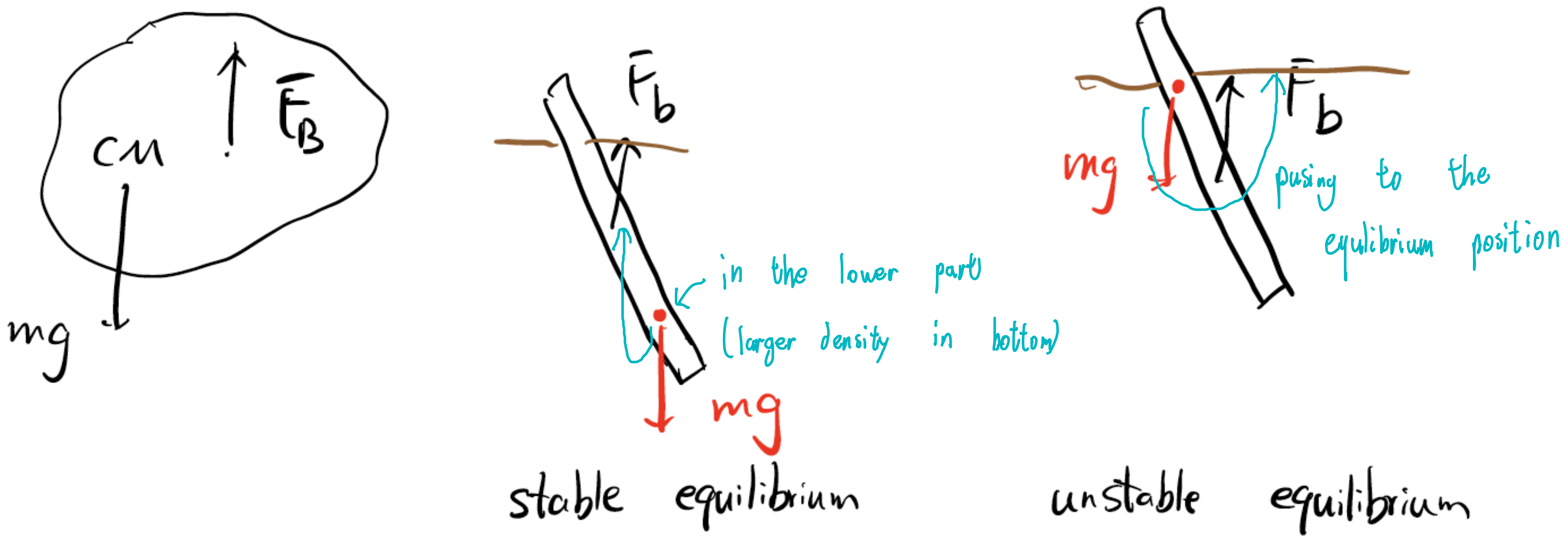
$V_{dis} = \frac{\rho_R}{\rho_w} \cdot V_R$

$> V_R$

② $V_{dis} = V_R$

water line $\downarrow$

Stability of Ships



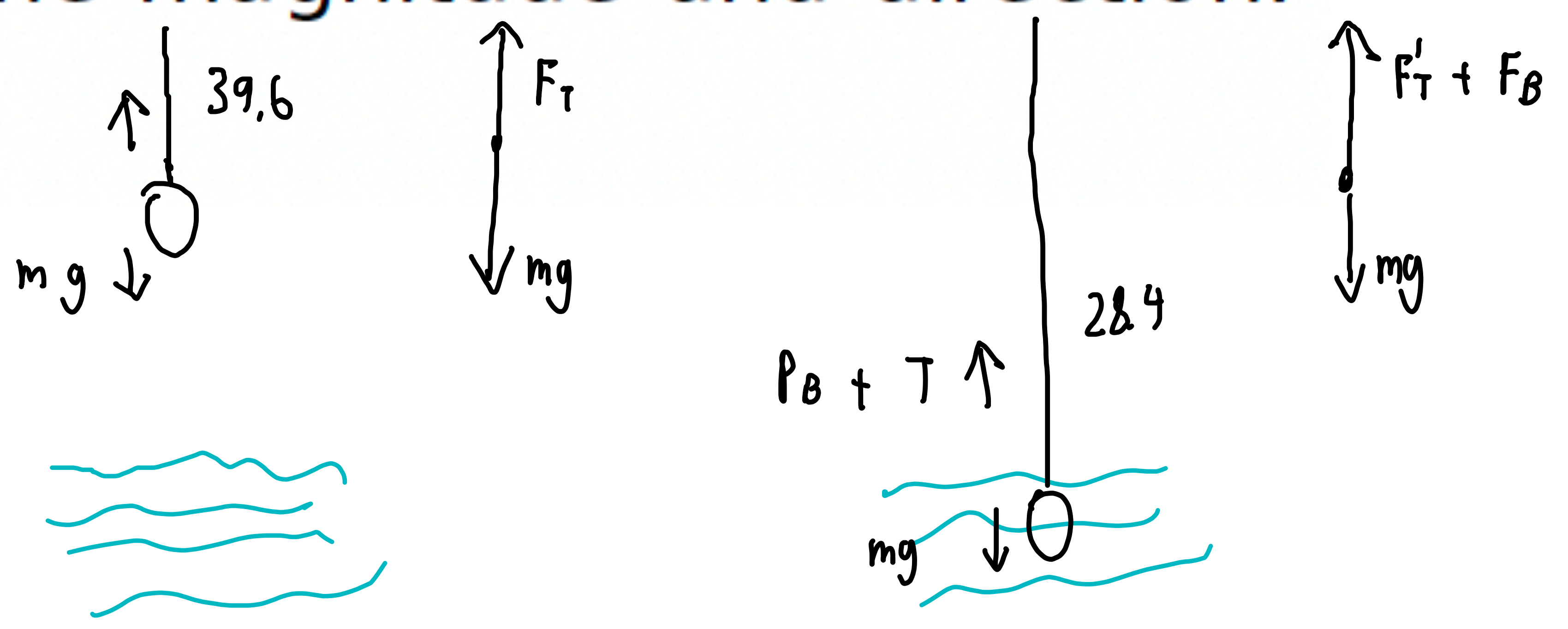
5. A rock is suspended by a light string. When the rock is suspended in the air at rest, the tension in the string is 39.6 N. When the rock is totally immersed in water, the tension is 28.4 N. (Hint: The density of water is $1.00\,\mathrm{g/cm^3}$. Neglect the buoyancy of the air. $g = 9.80\,\mathrm{m/s^2}$) **(10 pts)**

(a) What is the buoyancy of water on the rock? Indicate both the magnitude and direction. **(2 pts)**

Direction : upward

$$\begin{cases} F_T = mg \\ F_T' + F_B = mg \end{cases}$$

$$F_B = F_T - F_T' = 11.2\,N$$



(b) What is the reaction force of the rock against the water? Indicate both the magnitude and direction. **(2 pt)**

$$F_R = 11.2\,N$$

Direction : downward

(c) What is the volume of the rock? (Hint: Keep 3 significant figures.) **(3 pts)**

$$V_{rock} = V_{dis} = 1.14 \times 10^{-3}\ m^3 = 1.14\ L$$

$$F_B = \rho_s \cdot g \cdot V_{dis}$$

$$V = \frac{F_B}{\rho_s \cdot g}$$

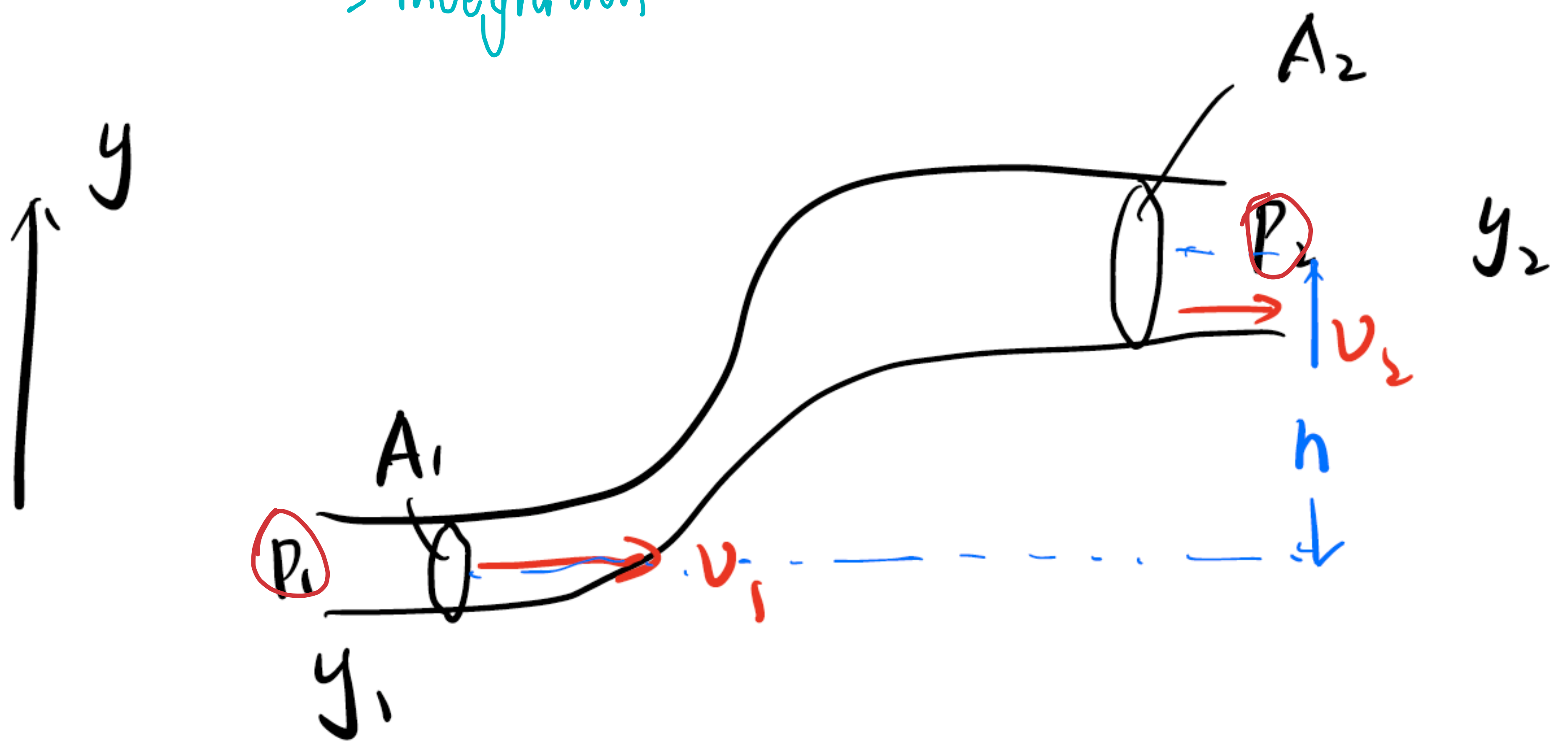
(d) What is the density of the rock? (Hint: Keep 3 significant figures.) **(3 pts)**

$$M = \rho_r V$$

$$\rho_r = \frac{38.6}{9.80 \cdot 1.14 \times 10^{-3}} = 3.54 \times 10^3\ kg/m^3$$

Fluid in Motion: Bernoulli's equation

→ integration



(before are static)

- Bernoulli's equation: $\frac{1}{2}\rho v^2 + \rho g h + P = \text{const}\,.$

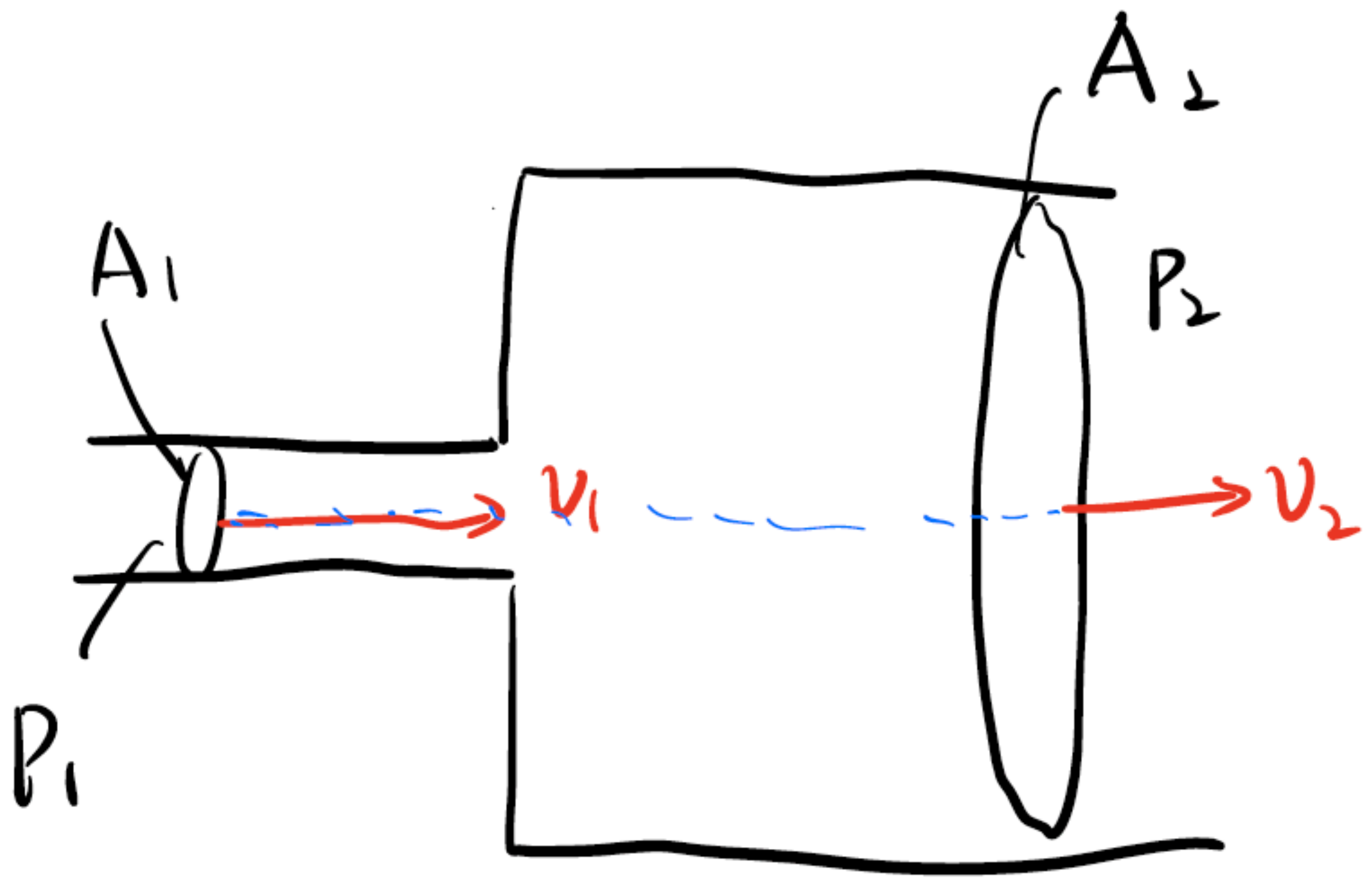
↳ pressure?

hystrostatic (static) : $P_2 - P_1 = \rho g (y_1 - y_2)$

$P_1 - P_2 = \rho g h$

fluid in motion (have KE)

$\frac{1}{2} m v^2 = KE$, $\frac{1}{2}\rho v^2 = \frac{KE}{V} + \frac{U}{V}$, $U = mgh$, $\frac{U}{V} = \rho g h$, $+ P = \text{const.}$



$$\frac{1}{2}\rho v_1^2 + \rho g h + P_1 = \frac{1}{2}\rho v_2^2 + \rho g h + P_2 \qquad (1)$$

continuity equation:

$$A_1 v_1 = A_2 v_2 \qquad (2)$$

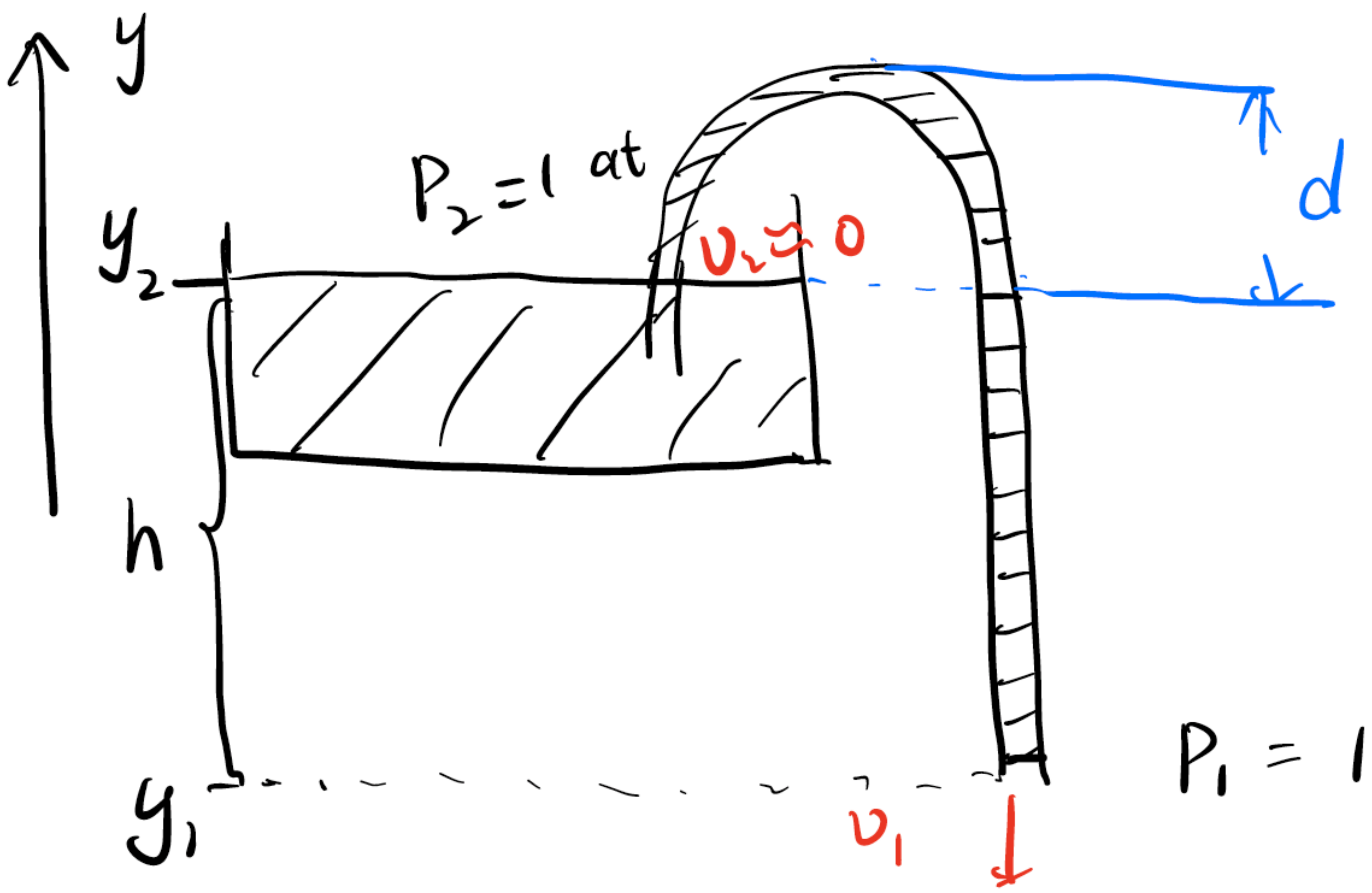
$(2) \to (1)$

$$\frac{1}{2}\rho\left(\frac{A_2}{A_1} v_2\right)^2 + P_1 = \frac{1}{2}\rho v_2^2 + P_2$$

$$P_2 - P_1 = \frac{1}{2}\rho v_2^2\left(\frac{A_2^2}{A_1^2} - 1\right) > 0$$

$$v_2 < v_1$$

Siphon

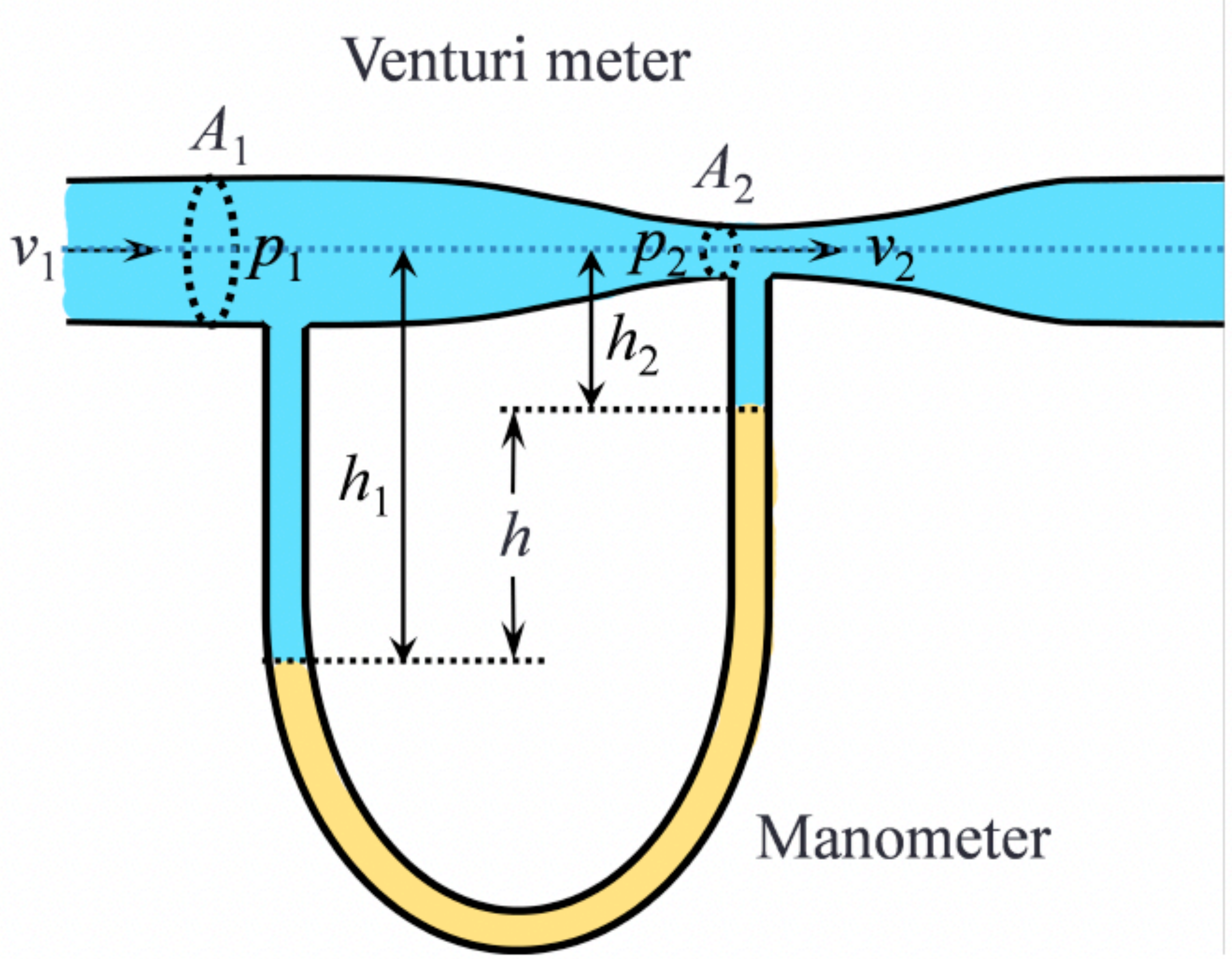


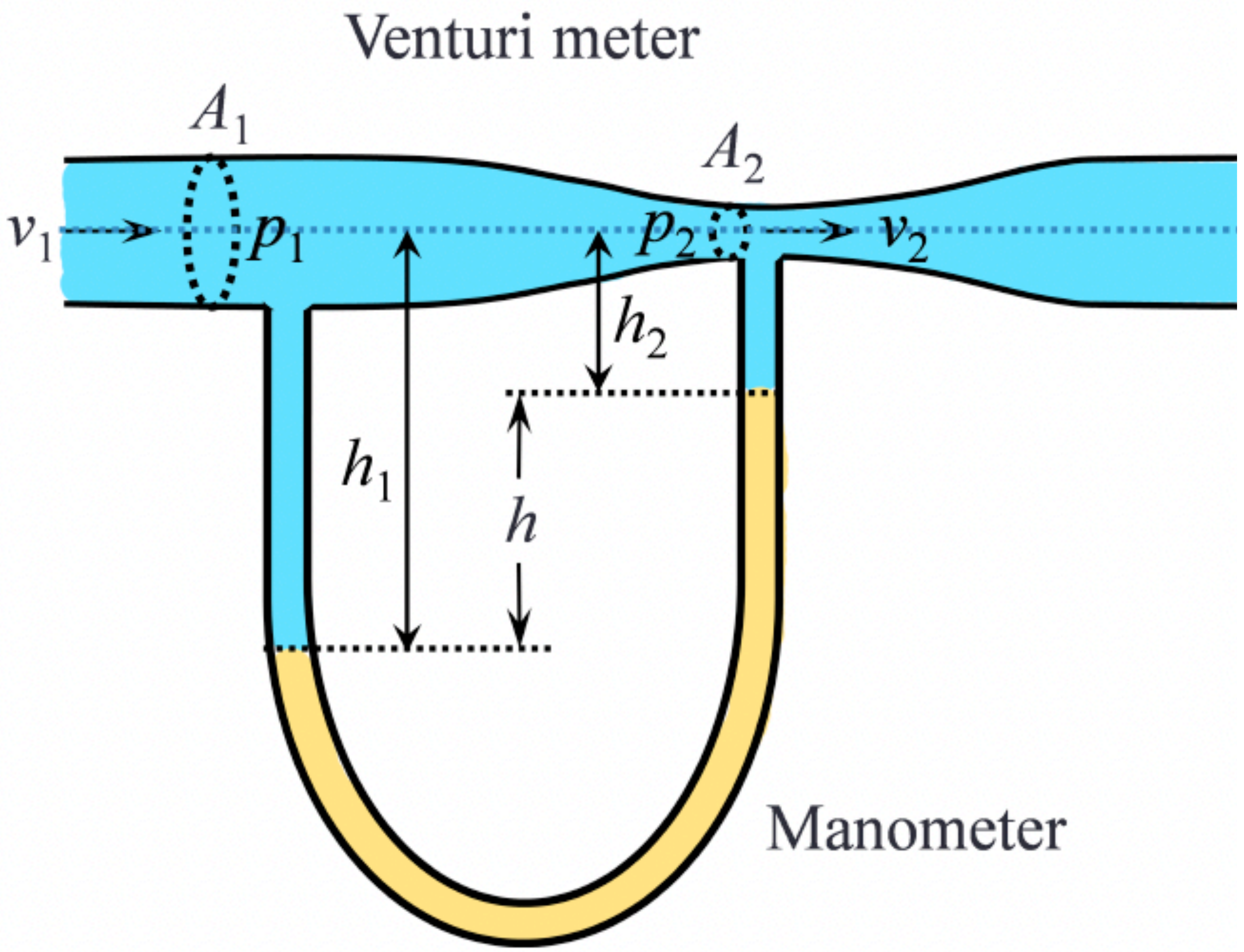
$$\frac{1}{2}\rho v_1^2 + \rho g h_1 + P_1 = \frac{1}{2}\rho v_2^2 + \rho g h_2 + P_2$$

$$\frac{1}{2}\rho v_1^2 = \rho g (h_2 - h_1)$$

$$v_1 = \sqrt{\frac{2gh}{\rho}}$$

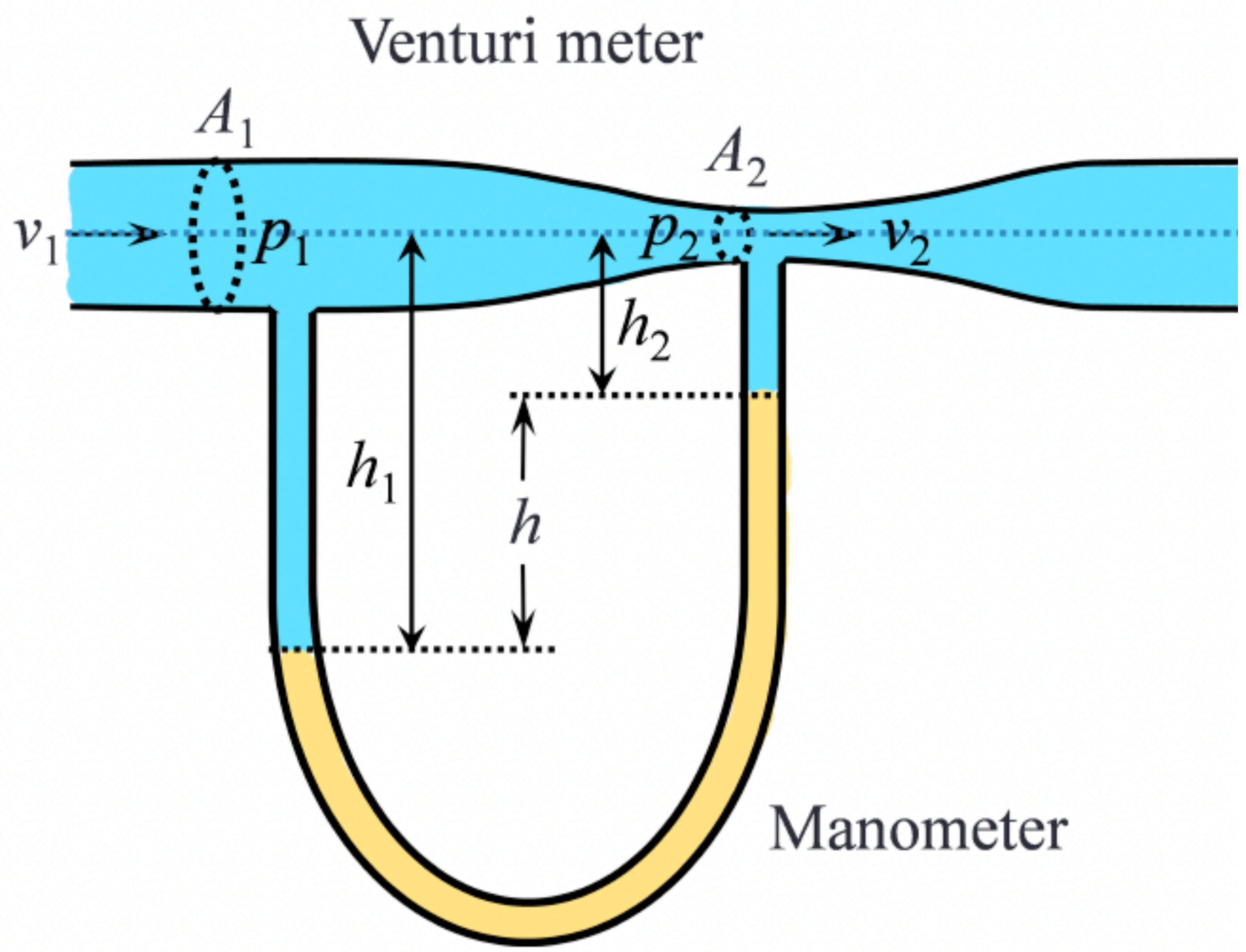
6. As shown below, the Venturi meter is connected to a U-shaped manometer. Consider an ideal fluid of density ρ_1 flowing through the horizontal pipe. The fluid at the portion of the pipe with a larger cross-section area A_1 has a flow speed v_1 and a measured pressure of p_1. The fluid at the portion of the pipe with a narrower cross-section area A_2 has a flow speed v_2 and a measured pressure of p_2. There is a steady flow in the horizontal pipe and the height difference of the fluid in the manometer remains constant at $h = h_1 - h_2$. **(10 pts)**





(a) Write down the continuity equation for the horizontal pipe.

(b) Use Bernoulli's equation to show $v_1 = \sqrt{\dfrac{2(p_1 - p_2)}{\rho_1[(A_1/A_2)^2 - 1]}}$



(c) Consider two different fluids of density ρ_1 (fluid in the horizontal pipe) and ρ_2 (fluid in the U-shaped manometer), where $\rho_1 < \rho_2$. Determine the flow speed v_1 in terms of ρ_1, ρ_2, A_1, A_2 and h, using part b. **(4 pts)**

5. In the figure below, the fresh water behind a reservoir dam has depth $D = 12$ m. A horizontal pipe passes through the dam at depth $d = 6.0$ m. The inner and outer parts of the pipe have cross-section areas of $S_1 = 18.8\text{cm}^2$ and $S_2 = 12.5\text{cm}^2$. A plug secures the pipe opening. **(10 pts)**

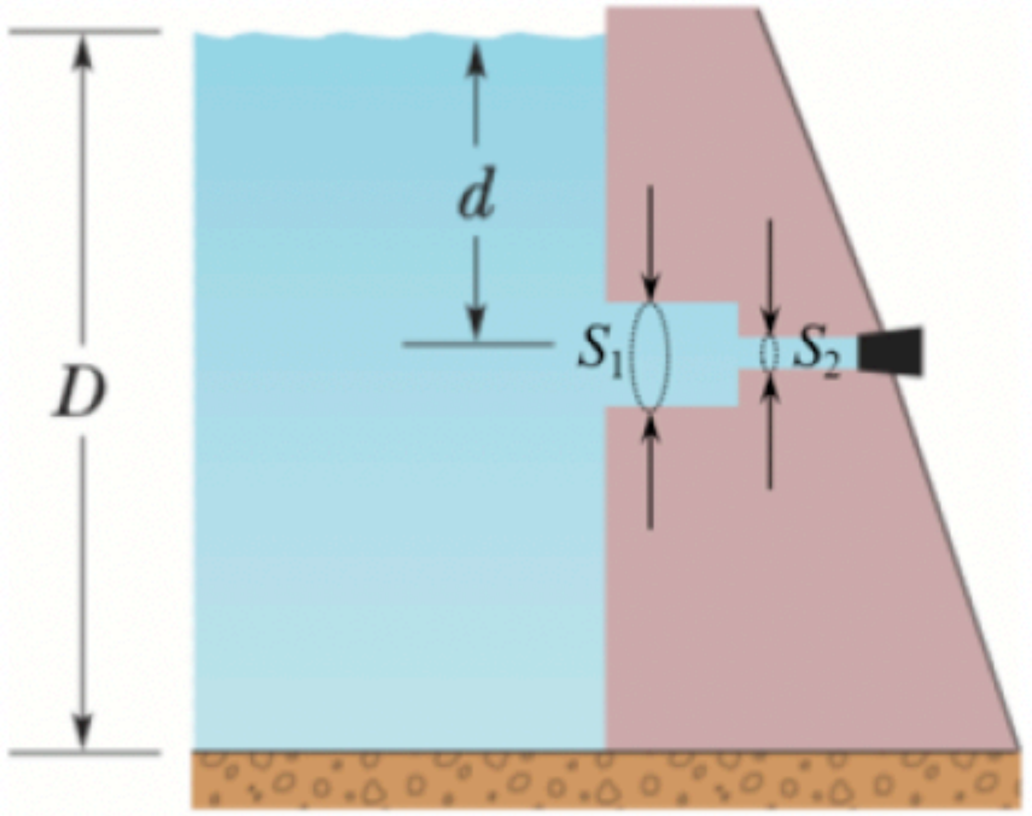
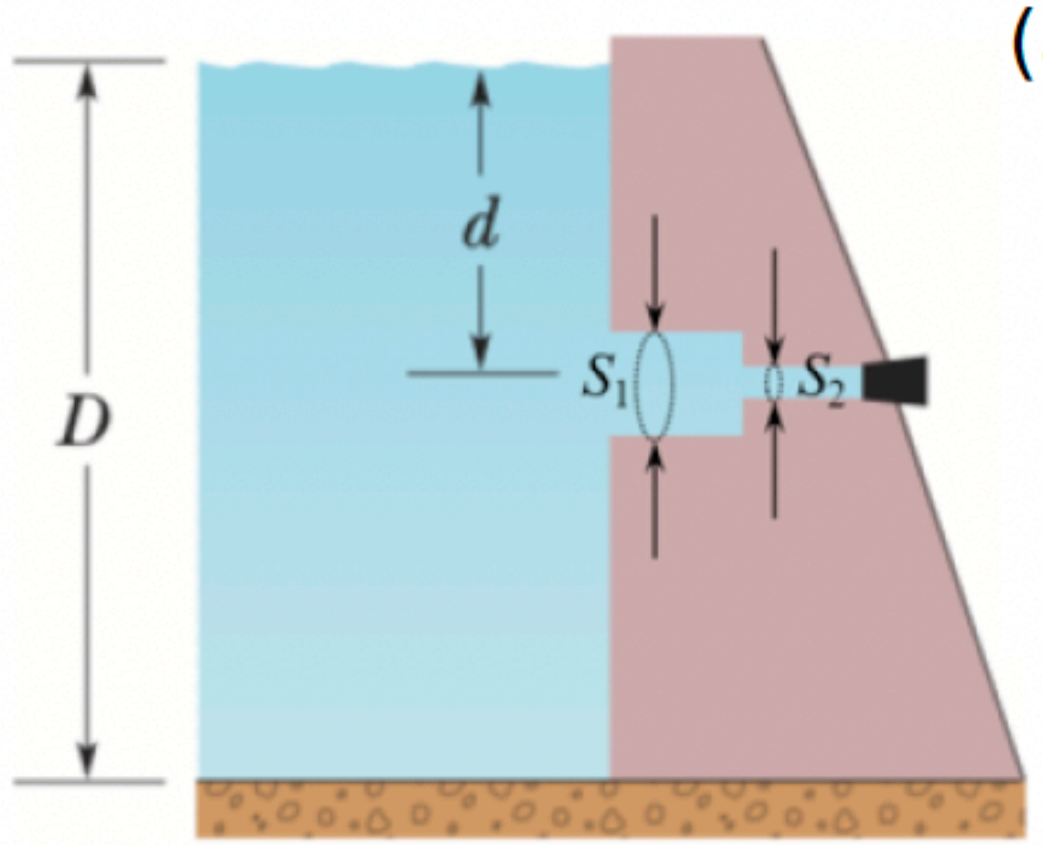


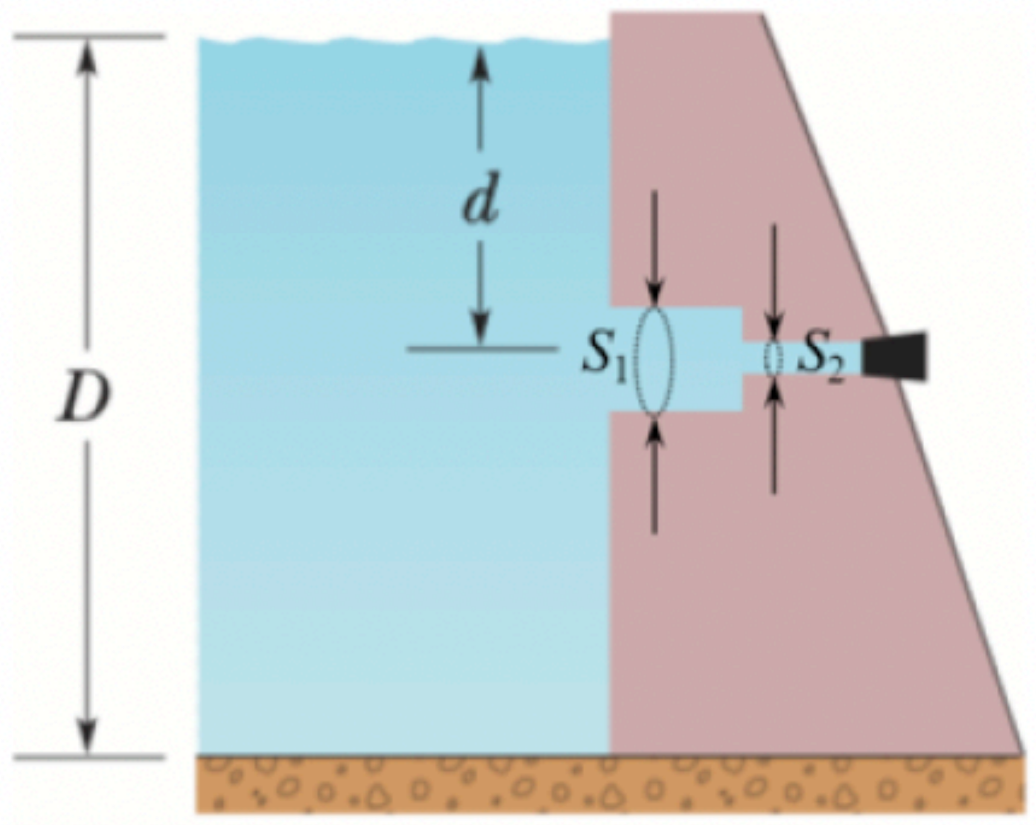
Fig. 4



(a) Find the values of gauge pressure in the horizontal pipe? **(2 pts)**

(Hint: The two diameters of the horizontal pipe are much smaller than d and D. The gauge pressure of water at the cross-section of S_1 approximately equals to that at the cross-section of S_2.) (The gauge pressure is defined as the difference between an absolute pressure (P) and the atmospheric pressure (P_{atm}).)

(b) Find the magnitude of the frictional force between plug and pipe wall. **(2 pts)**



(c) When the plug is removed, what is the water volume that exits the pipe in 1.0 hour? **(3 pts)**

(d) What is the speed of water flowing through the cross-section of S_1? **(3 pts)**